



UCLouvain

LINMA2222 - Stochastic Optimal Control & RL

Project - Part 4

Optimal Portfolio Strategy

Students

LUCAS AHO (35942200)

AYMERIC COUPLET (59482200)

Teachers

R. JUNGERS

G. BIANCHIN

G. BERGER

8 Mean-square Poisson error

Question 8.1

Compare the Poisson error with the Bellman error seen in the course for deterministic systems and for stochastic systems with discounted cost.

Answer

Given a policy π and approximations Q^θ and η^θ , the Poisson error is given by:

$$\mathcal{P}^\theta(x_t, u_t) := \mathbb{E}[r_t + Q^\theta(x_{t+1}, \pi(x_{t+1})) \mid x_t, u_t] - \eta^\theta - Q^\theta(x_t, u_t)$$

where r_t is the stage reward, η^θ is an approximation of the average reward per stage and $Q^\theta(\cdot, \cdot)$ is an approximation of the **relative** Q -function.

Alternatively, the Bellman error is given by:

$$\mathcal{B}^\theta(x_t, u_t) := \mathbb{E}[r_t + \gamma Q^\theta(x_{t+1}, \pi(x_{t+1})) \mid x_t, u_t] - Q^\theta(x_t, u_t)$$

with $\gamma \in [0, 1)$ the discount factor.

- **Deterministic Systems**

Given a deterministic system of the following form:

$$x_{t+1} = f(x_t, u_t)$$

The Poisson error (PE) is then given by:

$$\mathcal{P}^\theta(x_t, u_t) = r_t + Q^\theta(f(x_t, u_t), \pi(f(x_t, u_t))) - \eta^\theta - Q^\theta(x_t, u_t)$$

and the Bellman error (BE) by:

$$\mathcal{B}^\theta(x_t, u_t) = r_t + \gamma Q^\theta(f(x_t, u_t), \pi(f(x_t, u_t))) - Q^\theta(x_t, u_t)$$

The main differences are that PE has the $-\eta^\theta$ term and uses $\gamma = 1$.

- **Stochastic Systems**

For stochastic with discount γ , PE is **not** valid. In fact, the Poisson equation assumes an ergodic average-reward setting as well as $\gamma = 1$. In this case, BE must be used.

In the context of our project, we want to maximize the infinite-horizon average reward. Because PE measures how well our approximations Q^θ and η^θ are, this is the kind of problem where we'd want to use it instead of BE.

Question 8.2

Show that if Q^{θ_1} and η^{θ_2} are linearly parametrized (i.e., $Q^{\theta_1}(x, u) = \theta_1^\top \psi_Q(x, u)$ for some $\psi_Q : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^d$ and $\eta^{\theta_2} = \theta_2$), then minimizing the MSPE can be formulated as a linear system of equations to be solved in the least-square sense. Compute the matrix A_{lspc} and the vector b_{lspc} of the corresponding normal equations. Discuss the differences with the LSTD framework used in Part III.

Answer

In this setting, given a policy π , PE is given by:

$$\mathcal{P}^\theta(x, u) := \mathbb{E}[r + Q^{\theta_1}(x^+, \pi(x^+)) \mid x, u] - \eta_2^\theta - Q^{\theta_1}(x, u)$$

where $Q^{\theta_1}(x, u) = \theta_1^\top \psi_Q(x, u)$ and $\eta_2^\theta = \theta_2$.

Answer (cont.)

If we define the *expected next feature vector* as:

$$\bar{\psi}(x, u) := \mathbb{E}[\psi_Q(x^+, \pi(x^+)) \mid x, u]$$

and the expected reward as:

$$\bar{r}(x, u) := \mathbb{E}[r \mid x, u]$$

Then:

$$\mathbb{E}[r + Q^{\theta_1}(x^+, \pi(x^+)) \mid x, u] = \bar{r}(x, u) + \theta_1^\top \bar{\psi}(x, u)$$

and PE becomes:

$$\mathcal{P}^\theta(x, u) = \bar{r}(x, u) + \theta_1^\top \bar{\psi}(x, u) - \theta_2 - \theta_1^\top \psi_Q(x, u)$$

As suggested in the hint, we will define the following augmented vector:

$$\theta := [\theta_1 \ \theta_2]^\top$$

and define also:

$$\begin{aligned} \Psi(x, u) &:= \begin{bmatrix} \bar{\psi}(x, u) - \psi_Q(x, u) \\ -1 \end{bmatrix} \\ \implies \mathcal{P}^\theta(x, u) &= \bar{r}(x, u) + \theta^\top \Psi(x, u) \end{aligned}$$

We thus want to solve:

$$\begin{aligned} \min_{\theta} \mathbb{E}_{(x,u) \sim \mu} \left[(\bar{r}(x, u) + \theta^\top \Psi(x, u))^2 \right] &:= L(\theta) \\ \rightarrow \nabla_{\theta} L(\theta) = 2 \mathbb{E}_{(x,u) \sim \mu} [\Psi(x, u) (\bar{r}(x, u) + \theta^\top \Psi(x, u))] &= 0 \\ \iff \underbrace{\mathbb{E}_{(x,u) \sim \mu} [\Psi(x, u) \Psi^\top(x, u)]}_{A_{\text{lspc}}} \theta = \underbrace{-\mathbb{E}_{(x,u) \sim \mu} [\Psi(x, u) \bar{r}(x, u)]}_{b_{\text{lspc}}} \end{aligned}$$

The main difference with the previous LSTD framework is that we obtain an explicit estimation of the average reward η via θ_2 obtained by solving the the regression problem above.

Question 8.3

Propose an approximation of A and b in Question 8.2 from data obtained from the system controlled by some exploration policy π_{exp} , assuming that μ is the steady-state measure of the system controlled by π_{exp} . Then, explain how such a subroutine can be implemented if you have the transition kernel $\kappa(x' \mid x, u)$ of the system.

Answer

To approximate each quantity, we will first need to make ourselves a dataset. We will create the dataset as follows:

$$\{(x_i, u_i, r_i, x'_i)\}_{i=1}^N$$

where:

- $(x_i, u_i) \sim \mu$
- $x'_i \sim \kappa(\cdot \mid x_i, u_i)$
- r_i is the received reward

Answer (cont.)

To obtain the steady-state samples in practice, we run a long trajectory under π_{exp} , discards an initial period and take samples spaced apart to reduce correlation. Then, we have to be able to evaluate:

$$\bar{\psi}(x, u) = \mathbb{E}[\psi_Q(x^+, \pi(x^+)) \mid x, u]$$

We can do that under the stated assumption that we have access to a subroutine to compute:

$$\mathbb{E}[f(x^+, \pi(x^+)) \mid x, u] = \int f(x^+, \pi(x^+)) \kappa(x' \mid x, u) dx'$$

where f would be $\bar{\psi}$ in our case. To implement this subroutine, we can either compute it analytically if we are in the special case where κ has a closed-form expression for its primitive, or if it is not possible, we can use a Monte-Carlo approximation:

1. Given (x, u) , sample M next states $x'^{(1)}, \dots, x'^{(M)} \sim \kappa(\cdot \mid x, u)$
2. Compute:

$$\bar{\psi}_{\text{approx}}(x, u) = \frac{1}{M} \sum_{m=1}^M \psi_Q(x'^{(m)}, \pi(x'^{(m)}))$$

To compute an approximation of A and b , we need to do that for each sample (x_i, u_i) :

$$\longrightarrow \bar{\psi}_{\text{approx}}^{(i)} = \frac{1}{M} \sum_{m=1}^M \psi_Q(x_i'^{(m)}, \pi(x_i'^{(m)}))$$

We then construct $\hat{\Psi}^{(i)}$:

$$\longrightarrow \hat{\Psi}^{(i)} = \begin{pmatrix} \bar{\psi}_{\text{approx}}^{(i)} - \psi_Q(x_i, u_i) \\ -1 \end{pmatrix} \in \mathbb{R}^{d+1}$$

Finally, we can construct $\hat{A}$ and $\hat{b}$:

$$\hat{A} = \frac{1}{N} \sum_{i=1}^N \hat{\Psi}^{(i)} (\hat{\Psi}^{(i)})^\top$$

$$\hat{b} = -\frac{1}{N} \sum_{i=1}^N r_i \hat{\Psi}^{(i)}$$

where we assumed that we have access to r_i to act as an estimate for $\bar{r}(x_i, u_i)$.